

Evidence 4: Academic Publications & Global Research Impact

1. Research Output and Early Traction

As an independent researcher without the distribution channels of a traditional institution, I have produced four mathematically dense pre-print papers across open-access archives. For the purpose of demonstrating early research traction, the strongest evidence is concentrated in the first two papers shown in Figures 1 and 2 below. In less than two months, those two papers alone recorded a combined **271 views and 88 downloads**. For specialised deep-tech work published without institutional promotion, that level of early organic traction is notable.

The remaining two papers are newer outputs and I therefore present them primarily to show the continuity and breadth of the research programme, rather than to rely on them for the strongest traction argument.

2. The AMTTP Infrastructure Paper (TechRxiv)

Published: 27 February 2026 | **Metrics:** 152 Views, 49 Downloads (as of 6 April 2026)
DOI: 10.36227/techrxiv.177220113.33816607/v1

This inaugural paper introduced the architectural framework for bridging distributed ledger technology with off-chain ML inferences. Achieving 152 views and 49 direct downloads of a specialised technical document in roughly five weeks — without institutional promotion — is a strong early signal of interest in the applied systems engineering of the AMTTP protocol.

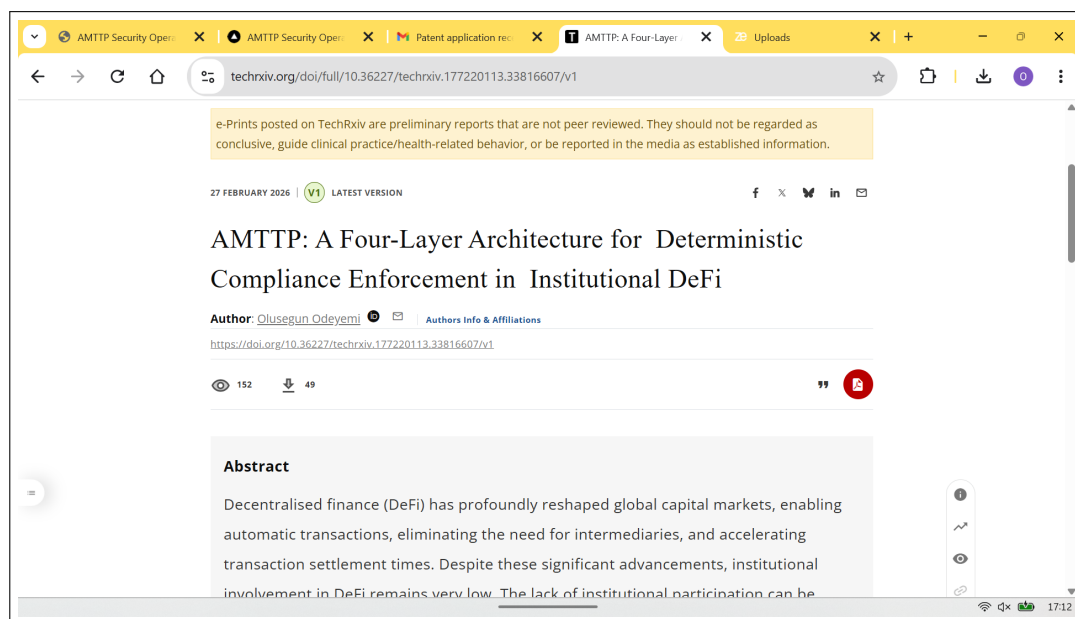


Figure 1: TechRxiv platform analytics showing 152 views and 49 downloads for the AMTTP infrastructure paper.

3. Blind-Spot Decomposition Theory — BSDT (Zenodo)

Published: 5 March 2026 | **Metrics:** 119 Views, 39 Downloads (as of 6 April 2026)
DOI: 10.5281/zenodo.18870590

Following the system architecture, I published the underlying mathematical models. BSDT proposes a geometric framework for decomposing machine learning fraud detection failures into four measurable components (Camouflage, Feature Gap, Activity Anomaly, Temporal Novelty). Achieving 119 views and 39 downloads within the same early publication window provides further credible evidence of interest in new theoretical approaches to ML blind-spot correction.

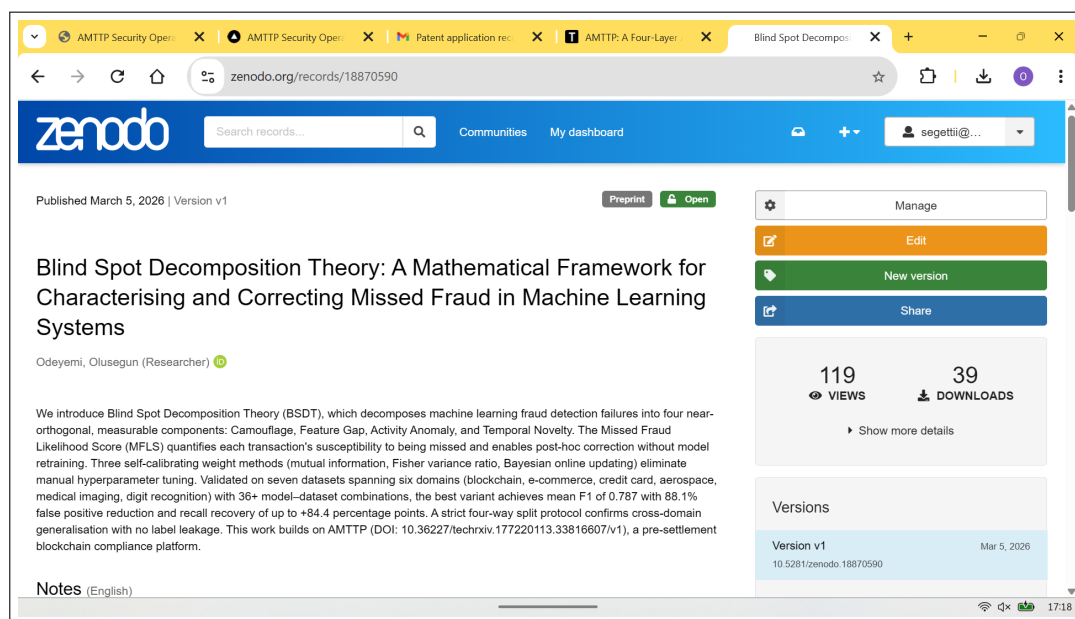


Figure 2: Zenodo platform analytics showing 119 views and 39 downloads for the BSDT mathematical framework.

4. Blind-Spot Decomposition and the Geometry of System Collapse (Zenodo)

Published: 16 March 2026 | **Metrics:** 10 Views, 16 Downloads (as of 6 April 2026)
DOI: 10.5281/zenodo.19038783

This third paper is, in my view, the strongest theoretical extension of the research programme. Rather than presenting BSDT only as a post-model fraud-analysis tool, it reformulates structural instability as a geometric and control problem. The paper shows that the BSDT energy functional is topologically equivalent to the interaction potential of a coupled dynamical system through Morse-index equivalence at critical points, two-sided gradient bounds, and homotopy equivalence of sublevel sets. It then derives a spectral phase-transition result: below a critical manifold, fixed coupling remains stable; above that manifold, no constant coupling is sufficient, and only adaptive friction achieves marginal stability.

That matters for this application because it demonstrates original mathematical work, not just implementation. The paper also connects the theory to concrete, non-trivial domains without retraining: FDIC banking data with a six-quarter lead before the 2008 GFC and AUROC 0.867 with zero false alarms in the documented benchmark window, Terra/Luna-style collapse dynamics with a 30-hour early alarm, and ERCOT Texas power-grid instability with AUC approaching 0.9996–0.9997 and a 72-hour lead before rolling blackouts. For Tech Nation purposes, I rely on this paper less for readership metrics and more as evidence that my work

has evolved from product architecture into reusable theory with potential relevance to regulated finance and critical infrastructure.

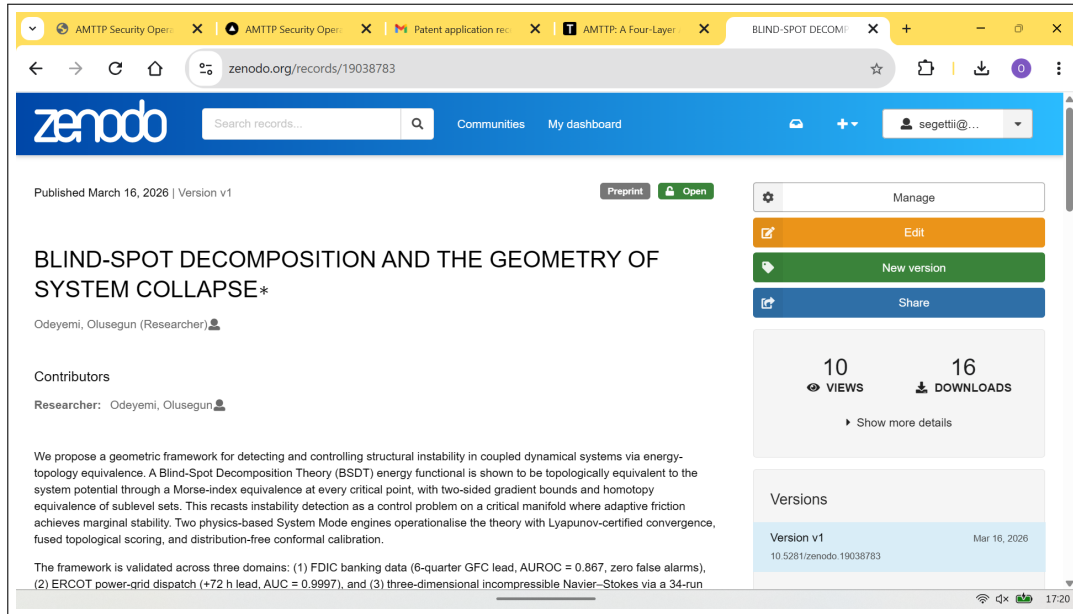


Figure 3: Zenodo platform analytics for the Geometry of System Collapse paper — positioning the BSDT framework as a reusable theory of instability across banking, crypto-collapse, and energy-system risk.

5. Universal Detection Principle (UDP)

DOI: 10.5281/zenodo.19037687

My fourth paper, **Universal Detection Principle (UDP)**, extends the research programme beyond post-model blind-spot correction into a broader multi-operator anomaly-detection framework. UDP formalises a tensor-style representation built from multiple mathematical operator families and explores how structurally different operators can be composed to improve anomaly detection, separability, and interpretability across datasets.

This paper is newer than the first two publications and I therefore do not rely on it for the strongest traction claim in this application. Instead, I include it because it shows the upward trajectory of the research arc: from AMTTP system architecture, to BSDT failure decomposition, to a more general theory of instability and control, and then to UDP as a broader operator-based detection framework.